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COMMUNICATION

Demonstration of hydrazide tagging for O-glycans and a central composite design of experiments optimization using the INLIGHT™ reagent

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Abstract

The INLIGHT™ strategy for *N*-linked glycan derivatization has been shown to overcome many of the challenges associated with glycan analysis. The hydrazide tag reacts efficiently with the glycans, increasing their non-polar surface area, allowing for reversed-phase separations and increased ionization efficiency. We have taken the INLIGHT™ strategy and adopted it for use with *O*-linked glycans. A central composite design was utilized to find optimized tagging conditions (45% acetic acid, 0.1 μg/μL tag concentration, 37 C, 1.75 h). Derivatization at optimized conditions was much quicker than any hydrazide derivatization strategy used previously. Human immunoglobulin A (IgA) and bovine submaxillary mucin (BSM) were then deglycosylated through hydrazinolysis and the removed glycans were tagged under optimum conditions. XIC of tagged glycans and MS2 data show successful hydrazide tagging of *O*-linked glycans for the first time.

Keywords O-glycosylation · Hydrazide tag · Rapid derivatization · Design of experiments

Introduction

Glycosylation is an important protein modification in biological systems. Glycans are involved in a myriad of roles, such as regulating protein function and protein-protein interactions within the cellular environment [1]. It is important to be able to develop analytical techniques to fully understand the role of glycans in biology. However, the study of *O*-linked glycans suffers from a few major challenges. Unlike *N*-linked glycans, *O*-glycans do not have a conserved core structure and an enzymatic strategy does not exist to remove all glycans from a protein [2]. *O*-glycans are attached to proteins on either serine or threonine residues and are composed of short, linear, or branched chains of saccharides with many possible isomers that can require robust separation techniques before analysis [3]. Glycans are highly polar molecules, which leads to poor ionization efficiencies when using electrospray ionization (ESI) mass spectrometry (MS) and prevents reversed-phase liquid chromatography (RPLC) separation without derivatization [4].

Previously we have shown the use of the Individuality Normalization when Labeling with Isotopic Glycan Hydrazide Tags (INLIGHT™) strategy to derivatize *N*-linked glycans for RPLC MS/MS analysis [5], including large-scale biological studies [6]. The INLIGHT™ strategy uses a hydrazide reagent with a large non-polar surface area (250 $Å^2$) that reacts near stoichiometrically with glycans in the non-reduced form. This non-polar surface area allows for reversed-phase separations and increases the electrospray ionization efficiency of the glycans. The incorporation of six ^{13}C stable isotope labeled sites also allows for relative quantification of paired samples.

While tagging techniques for *N*-glycans have proliferated, such as Tandem Mass Tags [7], the RapiFluor reagent [8], 2-AA [9] and 2-AB [10], INLIGHT™, and others, *O*-glycan chemical derivatization has received less attention. Due to their relative size, *O*-glycans are comparably less hydrophilic than *N*-glycans, although they still suffer from the

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hydrophobic bias in which hydrophobic species stay near the surface of a droplet and desorb from the droplet as the solvent evaporates, taking some of the surface charge with them. Additionally, current techniques do not allow for the analysis of freed *O*-glycans by positive-mode, RPLC ESI-MS/MS [4]. While *O*-glycopeptide analysis is possible and easier compared to *N*-glycopeptide analysis [11], the field has many challenges related to quantification of regions with *O*-glycan repeats resulting in data that is tedious and difficult to analyze. Thus, it may be desirable to look instead at global quantification changes of released *O*-glycans as a way of quantifying glycosylation changes on these targeted, problematic proteins, such as mucins [12]. Furthermore, while *O*-glycans should be able to react via the hydrazide reaction, we have been unable to document any reports of this reaction on the species across the glycan literature. Thus, application of INLIGHT™ for *O*-glycan derivatization offers opportunities to gain insights into carefully designed biological studies and to document a new chemistry.

Due to the diversity of *O*-linked glycan core structures, and their key structural differences to *N*-glycans [13] or glycosaminoglycans [14], INLIGHT™ tagging conditions must be optimized for many different potential structures. Instead of testing each variable one at a time, design of experiments (DOE) can be utilized to simultaneously optimize all variables [15]. A central composite design (CCD) is a full factorial design that also includes a center point and axial points for each variable [16]. This design works well when optimizing a few variables that have the potential for interaction and quadratic effects. Due to practical issues such as the time to complete each sample preparation, a DOE can be broken into blocks and performed over many days [17]. Multiple center points should be included in each block to account for potential random variability between experiments as well as to account for any systematic variation caused by running experiments over multiple days.

To deglycosylate an *O*-linked glycoprotein, chemical methods must be utilized [2]. One of the most common methods of chemical deglycosylation is through beta elimination, which removes the glycan using sodium hydroxide and sodium borohydride leaving the glycan in a reduced form [18]. Reduced glycans present an issue because the glycan can no longer be derivatized prior to analysis. The potential for a non-reductive beta elimination has been explored, however a non-specific method with minimal peeling, an unwanted side reaction in which the sugar at the reducing end degrades from the glycan, has not been developed [19–21]. Hydrazinolysis shows great potential as a non-specific deglycosylation method which leaves the glycans in non-reduced form, although it still suffers from the potential for unwanted peeling reactions [22]. Many studies have looked to reduce overall peeling during hydrazinolysis, the complete removal of water being the most effective method to date [23, 24].

In this study we will use a CCD to optimize the reaction conditions for hydrazide tagging of *O*-linked glycans using glycan standards. Trends in the variables effecting the hydrazide reaction efficiency will be explored in the context of the quadratic models produced. We will then validate these optimized conditions using hydrazinolysis to remove glycans from two standard glycoproteins, human immunoglobulin A (IgA) and bovine submaxillary mucin (BSM), followed by hydrazide tagging at optimized conditions using the INLIGHT™ reagent.

Experimental

Materials

The INLIGHT™ hydrazide reagents were obtained from Cambridge Isotope Laboratories, Inc. (Tewksbury, MA). All other reagents, including the three glycan standards (N-Acetyl-D-lactosamine, 3'-Sialyl-Lewis-a tetrasaccharide, and 4β-galactobiose) and two glycoproteins (human IgA and BSM) were obtained from Sigma Aldrich (St. Louis, MO). HPLC grade solvents (water, acetonitrile, and methanol) and glacial acetic acid were purchased from Fisher Scientific (Hampton, NH).

DOE optimization of INLIGHT™ *O*-glycan tagging

JMP Pro 12 (Cary, NC) was used to generate the central composite design for the optimization of tagging conditions. The reaction conditions were optimized for four variables, acetic acid (15–45%), temperature (37–60 °C), time (1.75–4.5 h), and concentration (0.10–0.25 μg/μL). The design was split into three blocks with two midpoints run in each block. A full list of experimental conditions can be found in Table 1. N-Acetyl-D-lactosamine (NAc), 3'-Sialyl-Lewis-a tetrasacharide (3Sia), and 4B-galactobiose (4β) (1 nmol concentrations) were tagged using the stable isotope labeled INLIGHT™ reagent at the various reaction conditions generated for the DOE. After addition of the required reagents, the samples were lightly vortexed and briefly centrifuged (~2 s) before being incubated for the required time. Afterwards the samples were briefly centrifuged and dried in a vacuum at 55 °C.

An internal standard to normalize run-to-run variability was generated by tagging the three glycan standards (10 nmol concentration) using the native INLIGHT™ reagent and following the *N*-glycan derivatization procedures described previously [5] (25% acetic acid, 0.25 μg/μL, 57 °C, 3 h). The tagged glycans were briefly centrifuged and dried in a vacuum at 55 °C.

Before analysis, the native and stable isotope labeled glycans were resuspended in 500 μL and 50 μL of HPLC grade H_2O respectively. The solution was pipetted up and down to

Table 1 A central composite design was used to evaluate the effect of four variables on O-glycan tagging efficiency

Block	Ranges tested	15–45	37–60	1.75–4.50	0.10–0.25
	Run	% acetic acid	Temperature (°C)	Time (hours)	Concentration (mg/mL)
1	1	45	37	1.75	0.10
	2	15	60	4.5	0.25
	3	30	48.5	3.125	0.175
	4	15	37	1.75	0.25
	5	15	60	1.75	0.10
	6	30	48.5	3.125	0.175
	7	45	60	4.5	0.10
	8	45	37	4.5	0.25
	9	45	60	1.75	0.25
	10	15	37	4.5	0.10
2	11	30	48.5	3.125	0.175
	12	15	37	4.5	0.25
	13	15	37	1.75	0.10
	14	45	37	1.75	0.25
	15	45	60	4.5	0.25
	16	15	60	1.75	0.25
	17	30	48.5	3.125	0.175
	18	45	60	1.75	0.10
	19	45	37	4.5	0.10
	20	15	60	4.5	0.10
3	21	30	48.5	3.125	0.175
	22	30	48.5	5.875	0.175
	23	30	71.5	3.125	0.175
	24	30	48.5	3.125	0.025
	25	30	48.5	3.125	0.325
	26	60	48.5	3.125	0.175
	27	0	48.5	3.125	0.175
	28	30	48.5	3.125	0.175
	29	30	48.5	0.375	0.175
	30	30	25.5	3.125	0.175

insure all glycans were suspended. After a brief vortex, the glycans were centrifuged at 13,000×*g* for 5 min and the supernatant was pipetted into a new vial to remove excess tag. For each experiment, 8 μL of spike in native standard and 8 μL of stable isotope labeled (SIL) glycans were combined before injection.

Hydrazinolysis of glycoproteins

The hydrazinolysis protocol (see Electronic Supplementary Material (ESM) Fig. S2) was adapted from a protocol generously provided by the Complex Carbohydrates Research Center (Personal Communication, Dr. Parastoo Azadi, 2017). In summary, 50 μg of glycoprotein was extensively dried (> 40 h) in a SpeedVac before addition of 100 μL of anhydrous hydrazine. The reaction was incubated at 60 °C for 6 h. The excess hydrazine was evaporated under a stream of nitrogen. To remove any additional hydrazine, the samples were washed twice with 250 μL of toluene and dried under nitrogen. The samples were re-acetylated using acetic anhydride in a 1.0 M sodium bicarbonate buffer. First 25 μL of acetic anhydride was added and incubated on ice for 10 min. Then an additional 25 μL of acetic anhydride was added and the samples were incubated at room temperature for an additional 50 min. To cleave any remaining hydrazones, 5 μL of 100 mM Cu (II) acetate in a 100 mM acetic acid buffer was added and allowed to react for 1 h before the samples were dried in a vacuum.

The samples were cleaned using a 300 mg Grace Alltech™ Extract-Clean™ SPE Carbograph SPE column (Columbia, Maryland) following the procedure developed by Packer et al. [25]. In short, the columns were washed with 20:80:0.1 H_2O:ACN:TFA (3 column volumes) followed by H_2O (3 column volumes). The samples were applied to the column and washed with H_2O (3 column volumes) before the glycans were eluted with 2 mL of 75:25:0.05 H_2O:ACN:TFA. The samples were then dried at 55 °C in a Savant SPD131 DDA SpeedVac Concentrator equipped with a Savant RVT5105 Refrigerated Vapor trap from Thermo Scientific (Waltham, MA).

Samples were tagged using the native INLIGHT™ reagent under optimized tagging conditions (45% acetic acid, 37 °C, 1.75 h, 0.10 μg/μL tag). After incubation, the samples were briefly centrifuged before drying in vacuum at 55 °C.

LC-MS analysis

Analysis was performed using an Easy nLC-1200 in-line with a Q-Exactive™ HF mass spectrometer (Thermo Scientific, San Jose, CA) 5 μL of sample was injected onto a 18 cm × 75 μm column packed with C18 2.6 μm, 100 Å resin. (Phenomenex, Torrance, CA) Mobile phase A was 98% H_2O, 2% acetonitrile, and 0.1% formic acid, while mobile phase B was 20% H_2O, 80% acetonitrile and 0.1% formic acid. The gradient went from 0% to 35% B in 2 min then to 55% B in 20 min before proceeding to 100% B in 4 min at a flow rate of 300 nL/min. The ionization conditions used were those previously optimized for glycan analysis [26]. The MS^1 scan parameters were 500 to 1500 *m/z* at a resolving power of $70{,}000_{FWHM}$ at *m/z* 200, a max injection time of 64 ms, AGC target of 5×10^5, 300 °C capillary temperature, and a spray voltage of 1.7 kV. The MS^2 conditions were a top 12 data dependent acquisition with a 15 s exclusion window at a resolving power of $15{,}000_{FWHM}$ at *m/z* 200, max injection time of 100 ms, AGC target of 5×10^4, 125 *m/z* fixed first mass, and a 10/20/30 stepped normalized collision energy.

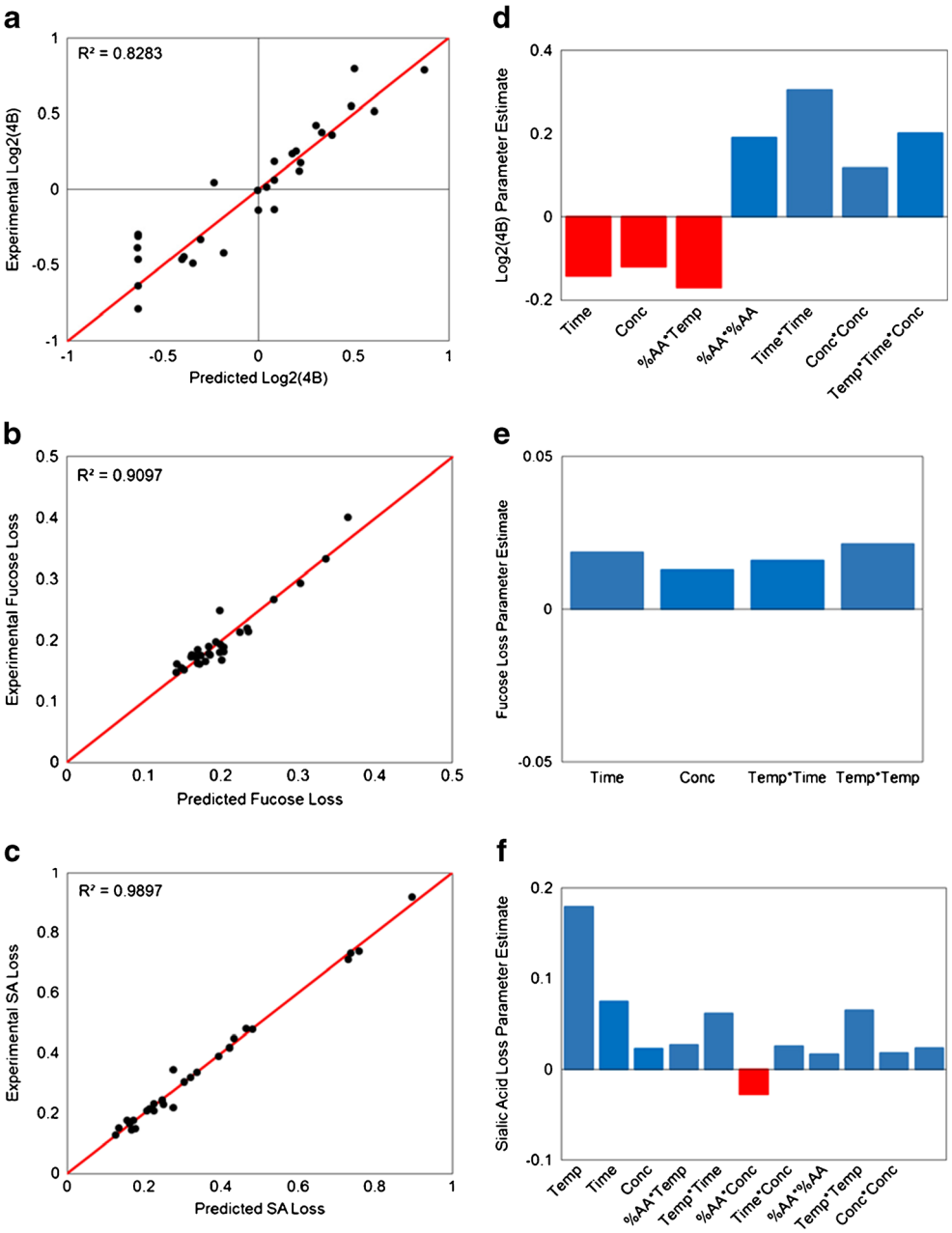


Fig. 1 Plots showing the experimental response versus the predicted response for **a** 4β-galactobiose, **b** Fucose loss, and **c** Sialic acid loss fit models. Parameter estimates of significant variables ($p < 0.05$) for the **d** 4β-galactobiose, **e** Fucose loss, and **f** Sialic acid loss models are also shown

Due to the comparably low molecular weights of *O*-linked glycans, SIL reagents with a 6 Da mass shift allowed for complete isotopic separation of the heavy and light tagged glycans. The results of the optimization study were therefore quantified by integrating the peak area of the monoisotopic peaks using the exact mass of each tagged glycan standard in Skyline [27] without correcting for the overlap of isotopic distributions found at high molecular weights needed in *N*-glycan analysis [5].

Results and discussion

For optimization, five responses were considered relevant, including maximization of the abundances of the standard glycans and minimization of the size peeling reactions of sialic acid or fucose. The abundance values for the intact glycans were normalized using an internal SIL standard to control for run-to-run variability, and this ratio was modeled for optimization (eq. 1). The response for labile sialic acid (Eq. 2) and fucose (Eq. 3) losses were calculated by taking the peak area of the peeled sugar glycan and dividing by the sum of the area of the intact glycan and peeled sugar glycan. Response surfaces were generated using a stepwise regression model in JMP Pro 12.0. Both the forward and reverse direction were tested using a minimized Bayesian information criterion (BIC) as the selection criteria and the direction that gave the highest R-squared value was chosen.

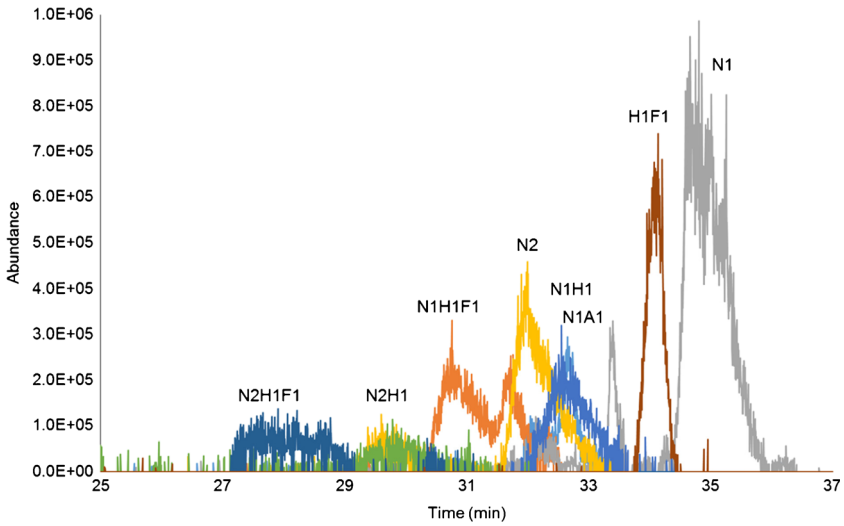


Fig. 2 XICs of native tagged *O*-glycans after removal from BSM. Chromatographic peaks are labeled with their corresponding glycan structure. (N—HexNAc, H—Hexose, F—Fucose, A—Neu5Ac)

$$Response = \frac{\sum \text{SIL } area}{\sum \text{NAT } area}, \text{per } glycan \quad (1)$$

$$Response = \frac{\sum \text{SIL}[3\text{Sia}{-}NeuAc]}{\sum \text{SIL}[3\text{Sia}{-}NeuAc] + \sum \text{SIL}[3\text{Sia}]} \quad (2)$$

$$Response = \frac{\sum \text{SIL}[3\text{Sia}{-}\text{Fuc}]}{\sum \text{SIL}[3\text{Sia}{-}\text{Fuc}] + \sum \text{SIL}[3\text{Sia}]} \quad (3)$$

Of the five models created, only the models for 4β-galactobiose, sialic acid loss (SA loss), and fucose loss were of excellent quality ($R^2 > 80\%$) (Fig. 1A–C) and ultimately used as the predictors to optimize the tagging response. In the 4β model, the quadratic effects were all significant and dominated the independent responses, such that increases in acetic acid, time, and concentration all resulted in increases in the overall tagging efficiency (Fig. 1D). The models were in good agreement with chemical principles, finding interactions between temperature, time, and concentration to also be a significant effect. While the model for the abundance of 3'-Sialyl-Lewis-a tetrasaccharide could not be fit ($R^2 < 0.5$), associated fits were generated on the peeling of fucose and sialic acid (Fig. 1B, C). Over the reaction conditions tested, sialic acid losses ranged from 12 to 90% and fucose losses ranged from 14 to 35%. For both labile sugar losses, the most significant effects, in regards to *p* value and magnitude, were temperature and time (Fig. 1E, F), in agreement with the literature on the factors promoting peeling [22]. Interestingly, the model for the abundance of N-Acetyl-D-lactosamine could also not be fit using a quadratic equation ($R^2 < 0.5$). The abundance values of NAc spanned a narrower range than that of 4β, and were not significantly affected by any of the reaction conditions. One of the rate limiting steps in the hydrazide reaction is the generation of the open-chain, reduced sugar, which is affected by the pH of the reaction solution and the localized chemical structure responsible for stabilizing the

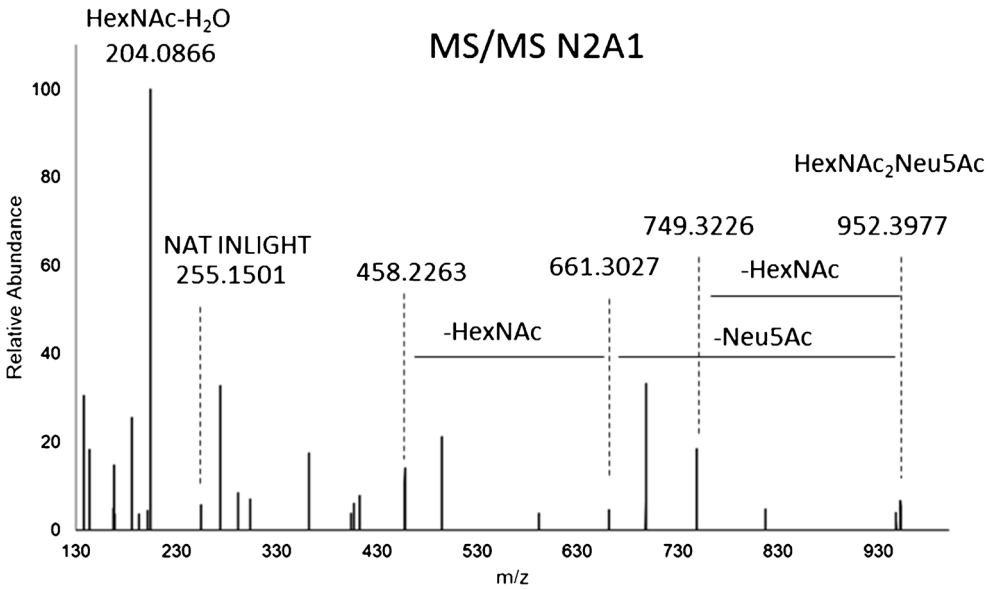


Fig. 3 MS/MS spectra of native tagged N2A1 from BSM. Fragmentation is consistent with a successfully native tagged *O*-glycan. Select peaks are labeled showing the fragments used for identification

aldehyde. Prior work has shown that the presence of an acetyl group substitution on C2 can stabilize the open-chain form [28], promoting this equilibrium under equivalent conditions compared to more simple sugars such as glucose and galactose. Galactose, the reducing sugar of 4β, is much more stable in the ring-structure, and thus may be more sensitive to reaction conditions.

Optimization of multiple predictor responses allows for the minimization of peeling and maximization of the abundances of the tagged glycans. As the factors promoting peeling also lead to increases in generation of the tagged species, a balance between the parameters is needed, and these complex interactions/response surfaces are shown in Fig. S1 in the ESM. The most optimal conditions were determined to be 45% acetic acid and 0.1 μg/μL INLIGHT™ tag incubated at 37 °C for 1.75 h. *O*-glycans are smaller on average than *N*-glycans and are therefore more linear and solvated in solution. Folded glycans are thought to react slower due to steric constraints, thus it is appropriate that the reaction conditions take approximately 50% of the incubation time, compared to *N*-glycans, and are completed at a comparably lower temperature (37 °C vs. 50 °C).

In order to validate the optimized tagging conditions, a full deglycosylation and tagging protocol was run on two standard glycoproteins known to contain *O*-linked glycans, human immunoglobulin A and bovine submaxillary mucin. Glycans were cleaved from the glycoprotein through hydrazinolysis and then tagged with native INLIGHT™ tag (ESM Fig. S2). The separation of INLIGHT™ *O*-glycans by RPLC was substantial, with 8 glycans eluting over a 10 min window (Fig. 2). Additionally, MS/MS confirmation of the glycans demonstrated that the generation of a peak for the INLIGHT™ tag can be used as a supplemental piece of information for identification (Fig. 3). A sample MS/MS spectra showing the Domon and Costello fragmentation of the native INLIGHT™ tagged glycan N2A1 is provided in Fig. S3 (see ESM). The diversity of glycans labeled with the native INLIGHT reagent showed for the first time that the hydrazide reaction is capable of reacting with many different *O*-glycan core structures.

Conclusion

In this study, we successfully extended the INLIGHT™ strategy for glycan derivatization to *O*-linked glycans. It was newly demonstrated that standard and hydrazinolysis-produced *O*-linked glycans can be labeled with a hydrazide reagent. Derivatization conditions were optimized to maximize the abundance of glycans and minimize peeling, resulting in an efficient protocol complete within 2 h. The applications, therefore, of this work may be readily extended to biological applications.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

References

1. Kudelka MR, Ju T, Heimburg-Molinaro J, Cummings RD. Simple sugars to complex disease—mucin-type O-glycans in cancer. Adv Cancer Res. 2015;126:53–135.
2. Zauner G, Kozak RP, Gardner RA, Fernandes DL, Deelder AM, Wuhrer M. Protein O-glycosylation analysis. Biol Chem. 2012;393:687–708.
3. Wuhrer M, Deelder AM, Hokke CH. Protein glycosylation analysis by liquid chromatography-mass spectrometry. J Chromatogr B. 2005;825:124–33.
4. Kailemia MJ, Ruhaak LR, Lebrilla CB, Amster IJ. Oligosaccharide analysis by mass spectrometry: a review of recent developments. Anal Chem. 2014;86:196–212.
5. Walker SH, Taylor AD, Muddiman DC. Individual normalization when labeling with isotopic glycan hydrazide tags (INLIGHT): a novel glycan relative quantification strategy. J Am Soc Mass Spectrom. 2013;24:1376–84.
6. Hecht ES, Scholl EH, Walker SH, Taykir AD, Cliby WA, Motsinger-Reif AA, et al. Relative quantification and higher-order modeling of the plasma glycan cancer burden ratio in ovarian cancer case-control samples. J Proteome Res. 2015;14:4394–401.
7. Yang S, Yuan W, Yang W, Zhou J, Harlan R, Edwards J, et al. Glycan analysis by isobaric aldehyde reactive tags and mass spectrometry. Anal Chem. 2013;85:8188–95.
8. Lauber MA, Yu YQ, Brousmiche DW, Hau Z, Koza SM, Magnelli P, et al. Rapid preparation of released N-glycans for HILIC analysis using labeling reagents that facilitates sensitive fluorescence and ESI-MS detection. Anal Chem. 2015;87:5401–9.
9. Bigge JC, Patel TP, Bruce JA, Goulding PN, Charles SM, Parekh RB. Nonselective and efficient fluorescent labeling of glycans using 2-amino benzamide and anthranilic acid. Anal Biochem. 1995;230:229–38.
10. Qian J, Liu T, Yang L, Daus A, Crowley R, Zhou Q. Structural characterization of N-linked oligosaccharides on monoclonal antibody cetuximab by the combination of orthogonal matrix-assisted laser desportion/ionization hybrid quadrupole-quadrupole time-of-flight tandem mass spectrometry and sequential enzymatic digestion. Anal Biochem. 2007;364:8–18.
11. Desaire H. Glycopeptide analysis, recent developments and applications. Mol Cell Proteomics. 2013;12:893–901.
12. Burchell JM, Mungul A, Taylor-Papadimitriou J. O-linked glycosylation in the mammary gland: changes that occur during malignancy. J Mammary Gland Biol Neoplasia. 2001;6:355–64.
13. Walker SH, Budhathoki-Uprety J, Novak BM, Muddiman DC. Stable-isotope labeled hydrophobic hydrazide reagents for the relative quantification of N-linked glycans by electrospray ionization mass spectrometry. Anal Chem. 2011;83:6738–45.
14. Mangrum JB, Mehta AY, Alabbas AB, Desai UR, Hawkridge AM. Comparative analysis of INLIGHT™-labeled enzymatically depolymerized heparin by reverse-phase chromatography and

high-performance mass spectrometry. Anal Bioanal Chem. 2017;409:499–509.
15. Hecht ES, Oberg AL, Muddiman DC. Optimizing mass spectrometry analyses: a tailored review on the utility of design of experiments. J Am Soc Mass Spectrom. 2016;27:767–85.
16. Box GEP, Wilson KB. On the experimental attainment of optimum conditions. J R Stat Soc. 1951;1:1–45.
17. Fisher RA. An examination of the different possible solutions of a problem in incomplete blocks. Ann Eugenics. 1940;10:52–75.
18. Iyer RN, Carlson DM. Alkaline borohydride degradation of blood group H substance. Arch Biochem Biophys. 1971;142:101–5.
19. Yu G, Zhang Y, Zhang Z, Song L, Wang P, Chai W. Effects and limitation of excess ammonium on the release of O-glycans in reducing forms from glycoproteins under mild alkaline conditions for glycomic and functional analysis. Anal Chem. 2010;82:9534–42.
20. Goetz JA, Novotny MV, Mechref Y. Enzymatic/chemical release of O-glycans allowing MS analysis at high sensitivity. Anal Chem. 2009;81:9546–52.
21. Haung Y, Mechreg Y, Novotony MV. Microscale nonreductive release of O-linked glycans for subsequent analysis through MALDI mass spectrometry and capillary electrophoresis. Anal Chem. 2001;73:6063–9.
22. Patel T, Bruce J, Merry A, Bigge C, Wormald M, Jaques A, et al. Use of hydrazine to release in intact and unreduced form both N- and O-linked oligosaccharides from glycoproteins. Biochemistry US. 1993;32:679–93.
23. Merry AH, Neville DC, Royle L, Matthews B, Harvey DJ, Dwek RA, et al. Recovery of intact 2-aminobenzamide-labeled O-glycans released from glycoproteins by hydrazinolysis. Anal Biochem. 2002;304:91–9.
24. Hecht ES, McCord JP, Muddiman DC. Definitive screening design optimization of mass spectrometry parameters for sensitive comparison of filter and SPE purified, INLIGHT plasma N-glycans. Anal Chem. 2015;87:7305–12.
25. Kozak RP, Royle L, Gardner RA, Bondt A, Fernandes DI, Wuhrer M. Improved nonreductive O-glycan release by hydrazinolysis with ethylenediaminetetraacetic acid addition. Anal Biochem. 2014;453:29–37.
26. Packer NH, Lawsom MA, Jardine DR, Redmond JW. A general approach to desalting oligosaccharides released from glycoproteins. Glycoconj J. 1998;15:737–47.
27. Loziuk PL, Hecht ES, Muddiman DC. N-linked glycosite profiling and use of skyline as a platform for characterization and relative quantification of glycans in differentiating xylem of *Populus trichocarpa*. Anal Bioanal Chem. 2017;409:487–97.
28. Kilcoyne M, Gerlach JQ, Farrell MP, Bhavanandan VP, Joshi L. Periodic acid-Schiff's reagent assay for carbohydrates in a microtiter plate format. Anal Biochem. 2011;416:18–26.